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Indian Institute of Technology Bhubaneswar

School of Electrical Sciences

Mid Semester (Spring) Examination – 2025

Subject Name: Natural Language Processing
Date: 24/02/2025
Duration: 2 Hours

Subject Code: CS6L027
Time: 3:00 P.M. - 5:00 P.M.
Full Marks: 50

Instructions

- Write answers as precisely as possible. Stories will negatively affect your marks.
- Make necessary and appropriate assumptions. Write your assumptions clearly before answering a question.
- No notebooks are allowed.

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1. Multiple choice questions [There could be multiple correct options, no partial marking] **(5 * 1)**

   1.1. N-fold cross-validation is a technique per round that uses
   a. All data in the corpus for training
   b. (100/N) % of the data in the corpus for training
   c. 90% of the data in the corpus for training
   d. ((N-1)*(100/N))% of the data in the corpus for training

   1.2. *'I **never** said she stole my money'* - The stress is on the **bold** word in this sentence. What does the sentence intend to mean
   a. I simply didn't ever say it.
   b. I might have implied it in some way, but I never explicitly said it.
   c. I said she stole someone else's money.
   d. I just said she probably borrowed it.

4
1
3
2
2

1.3. The length of the stem in the word '**uncommunicativeness**' is

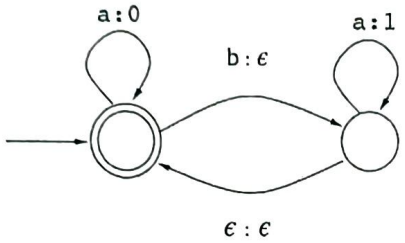
a. 10 b. 12 c. 13 d. None of the above

1.4. ***'I saw someone on the hill with a telescope'*** - is an example of which type of ambiguity

a. Lexical ambiguity b. Semantic ambiguity

c. Syntactic ambiguity d. Pragmatic ambiguity

1.5. What output/s will this FST produce, given the input **aababa**?



a. 0000 b. 0011 c. 1000 d. 0001

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2. Point out why the class of regular grammar and context-free grammar are not expressive enough to model natural language. **(4)**

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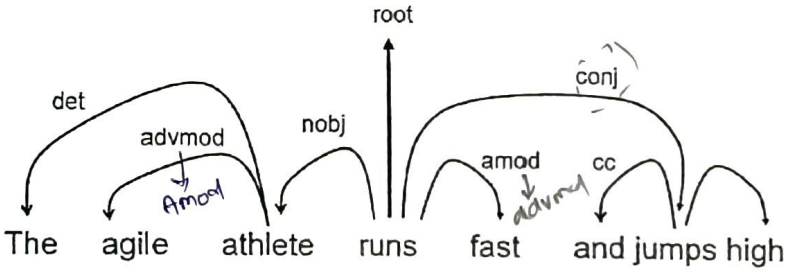
3. How many things are wrong in the following dependency parse tree? **(4)**



Hints: det → determiner, advmod → adverbial modifier, nobj → nominal object, Amod → adjective modifier, conj → conjunct, cc → coordinating conjunction

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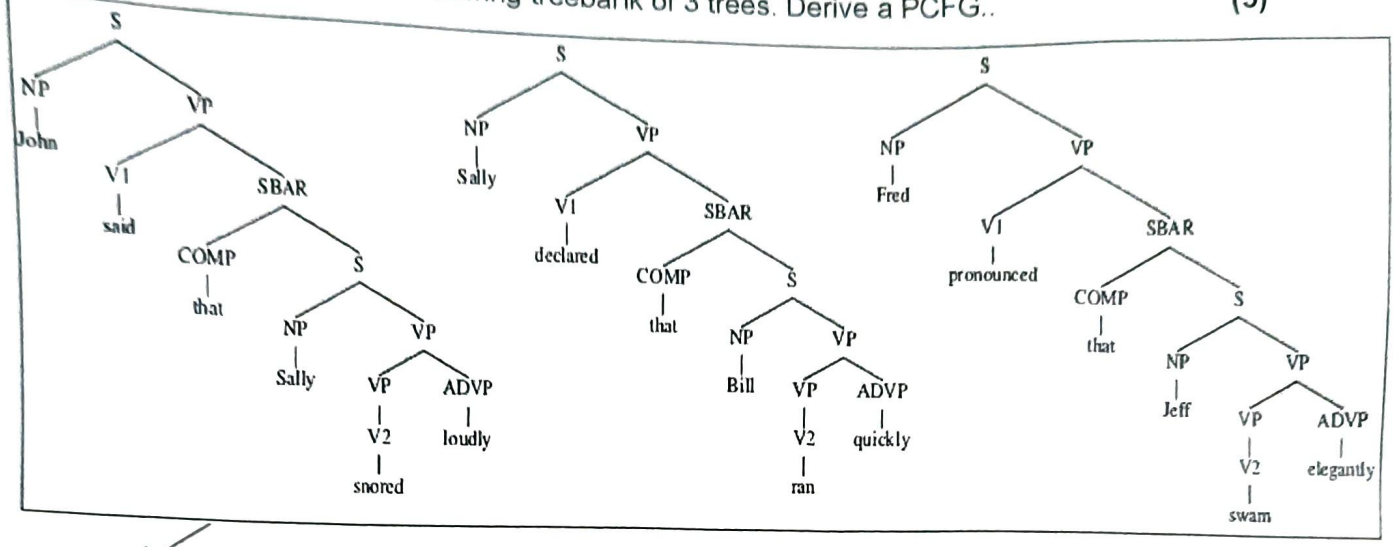
4.

4.1. Consider the following treebank of 3 trees. Derive a PCFG.. **(5)**



4.2. Show the possible parse trees for the following sentence using the PCFG constructed in 4.1 and calculate their probabilities using the CYK algorithm. **(8)**

Jeff pronounced that Fred snored loudly

4.3. Is the higher probability tree the right one? Justify. **(2)**

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5. Train a Parts-of-speech Hidden Markov Model (HMM) with Maximum Likelihood Estimation using the following annotated text (three sentences). Draw the HMM. **(6)**

the/D cat/N can/VA fish/VV a/D fish/N
the/D fish/N is/VV in/P the/D fish/N can/N
workers/N can/VV the/D fish/N

*** You can assume <s> and </s> as the start and end of the sentence, respectively.

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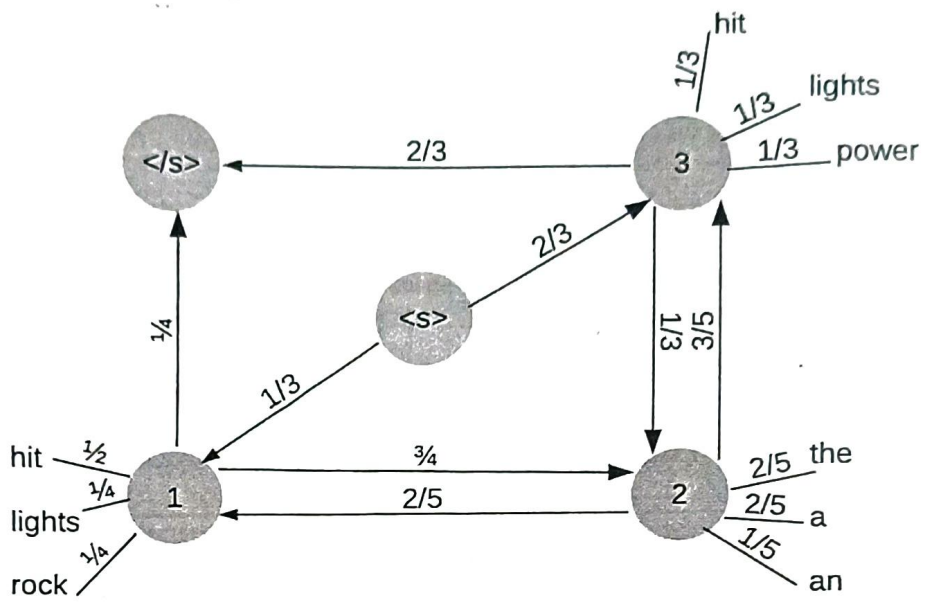
6. What is the most probable label (labels are 1,2,3) sequence and its probability on the sentence below, given the following HMM? **(8)**

<s> hit the lights </s>



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7. Draw a transducer for the K-insertion rule (intermediate form → surface form) **(8)**

The rule is → ***Words ending with vowel + c, add k before the suffix -ed***

For example: *panic ^ ed # → panicked*

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dis~~tu~~rbed

decided

dec

distrubed